

Polynomial

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1 Introduction to Polynomial

1.1 Field

Let F be the set of some numbers. If F is closed to addition, subtraction, multiplication and division, then F is called a field.

For example: rational set $\mathbb{Q}$ (Quotient) $\rightarrow$ rational field

1.2 Polynomial

1.2.1 Definition

For a field F and a notation(variation) X (probably not $\in F$),

$$f(X) = a_n X^n + a_{n-1} X^{n-1} + \dots + a_1 X + a_0 \text{ where } a_i \in F, i = 1, 2, \dots, n$$

is a polynomial with single variation on F .

1.2.2 Basic Features

1. If $n \geq 0$ and $a_n \neq 0$, then $n = \deg f(x)$.
2. Any $a \neq 0 \in F$ can be seen as a polynomial with degree 0.
3. $0 \in F$ is a polynomial with degree $-\infty$ and with no first term.
4. For polynomials $f(X) = a_n X^n + \dots + a_0$ and $g(X) = b_m X^m + \dots + b_0$,

$$f(X) = g(x) \Leftrightarrow m = n \text{ and } a_i = b_i, i = 1, 2, \dots, n$$

5. $\deg fg = \deg f + \deg g$
6. $f \neq 0, g \neq 0 \Leftrightarrow fg \neq 0 \Leftrightarrow F[X]$ has no null factor.
7. Most polynomials $f(X)$ on F are non-invertible. The only invertible polynomial is non-zero constants(F^*).

2 Number Theory in Polynomial

2.1 Euclidean Division

$\forall f, g \in F[X]$ and $g \neq 0$, $\exists! q, r \in F[X]$ such that

$$f = gq + r \quad (r = 0 \text{ or } \deg r < \deg g)$$

2.2 Factors of $f(X)$

2.2.1 Factor

If $f = gq$ ($g \neq 0$), then $g|f$ so g is a factor of f .

Trivial factors: 1 and $f(X)$ itself.

2.2.2 Reducibility

Also, if $f = gq$ and $g, q \in F$ are not constants, then f is reducible over F .

If $f(X)$ is irreducible (i.e. only has trivial factors), then f is prime.

2.2.3 Association

If $g|f$, then $bg|af$ for any non-zero constant $a, b \in F$.

For any non-zero constant $c \in F$, g and cg are associate.

If $g|f$ and $f|g$, then f and g are associate.

2.2.4 Highest Common Factor

$$\begin{aligned} d(X) \text{ is the HCF of } f(X) \text{ and } g(X) \\ \Updownarrow \\ d|f, d|g \text{ and } \forall c \in F, c|f, c|g, c|d \end{aligned}$$

The HCF of f and g is unique if associate is not considered.

If $(f, g) = 1$, then f and g are coprime.

2.3 Bézout Identity

For $f(X)$ and $g(X) \in F[X]$, their HCF $d(X)$ always exists uniquely and there exists $u(X), v(X) \in F[X]$ such that

$$u(X)f(X) + v(X)g(X) = d(X)$$

Particularly,

$$\begin{aligned} f(X) \text{ and } g(X) \text{ are coprime} \\ \Updownarrow \\ \exists u(X), v(X) \in F[X] \text{ such that } u(X)f(X) + v(X)g(X) = 1 \end{aligned}$$

Generally,

For $f_i \in F[X]$, $i = 1, 2, \dots, s$, $\exists u_i \in F[X]$, $i = 1, 2, \dots, s$ such that

$$u_1 f_1 + u_2 f_2 + \dots + u_s f_s = (f_1, f_2, \dots, f_s)$$

2.4 Unique Factorisation Domain

$\forall$ non-constant $f(X) \in F[X]$ can be factorised uniquely:

$$f(X) = p_1(X)^{v_1} p_2(X)^{v_2} \dots p_r(X)^{v_r} \text{ where } p_i(X) \text{ are prime polynomials in } F[X].$$

$$p_i(X)^{v_i} || f(X)$$

This theorem can be stated as $F[X]$ is a unique factorisation domain.

2.5 Remainder Theorem and Factor Theorem

Remainder Theorem: $f(X)$ has remainder $r = f(c)$ when divided by $(X - c)$

Factor Theorem: $(X - c) | f(X) \Leftrightarrow f(c) = 0$

If $f(X) = (X - c)^m g(X)$ ($g(c) \neq 0, m \geq 1$), c is called multiple root when $m \geq 2$, c is called single root when $m = 1$.

2.6 Fundamental Theorem of Algebra

Theorem: If $f(X)$ is of degree n ($n \geq 0$) in $F[X]$, then $f(X)$ has at most n roots over F .

(Proof by induction)

Attention: If we deal with a set which is not a field, then a polynomial with degree n could have $> n$ roots.

Corollary: If $f, g \in F[X]$ are both of degree $\leq n$ ($n \geq 0$) and have $n + 1$ points with equal values, then $f = g$.

(Proof: consider $h = f - g$)

2.7 Differentiation

To justify the multiple roots of polynomial, we introduce the concept of differentiation.

2.7.1 Definition

The differentiation form of $f(X) = a_n X^n + a_{n-1} X^{n-1} + \dots + a_1 X + a_0$ is

$$f'(X) = n a_n X^{n-1} + (n-1) a_{n-1} X^{n-2} + \dots + a_1 \in F[X]$$

2.7.2 Theorem

For $f(X) \in F[X]$, $c \in F$,

Theorem: c is the root of $f(X) \Leftrightarrow f(c) = f'(c) = 0 \Leftrightarrow c$ is the root of (f, f')

Corollary1: If $(f, f') = 1$, then $f(X)$ has no multiple root in any field $E \supseteq F$

Corollary2: If $f(X)$ is irreducible in $F \subset \mathbb{C}$, then $f(X)$ has no multiple root.

2.8 Summary

The application of number theory in polynomial is part of algebraic number theory.